UAV Slung-Payload SITL: Technical Evaluation Report

Kinematic Aggressiveness vs. Bandwidth Interpretation & Environment Reproducibility

Author: Bathala Paneendra Kumar (NIT Durgapur)

Recipient: Prof. Alessandro Zavoli & Fabio Polese

Project: Autonomous UAV with Slung Payload

Affiliation: Sapienza - University of Rome / NIT Durgapur

Repository: github.com/paneendrakumar0/uav-autonomous-telemetry

Date: October 11, 2026

1. Executive Summary

This technical report addresses the constructive feedback provided by **Professor Alessandro Zavoli** regarding the UAV Slung-Payload Simulation-in-the-Loop (SITL) project. Specifically, this document:

- Calculates the analytical kinematic characteristics (peak velocity v_{max} and peak acceleration a_{max}) of the commanded reference trajectory.
- Distinguishes whether tracking degradation across trajectory rates is driven by *trajectory aggressiveness* or *controller bandwidth / tuning*.
- Presents the containerized Docker and VS Code DevContainer architecture ensuring 1-command reproducibility for student researchers.
- Documents the automated reference test case with empirical tolerances and metrics.

2. Kinematic Aggressiveness vs. Controller Bandwidth

The reference Figure-8 path is parameterized in the horizontal plane with amplitude $A = 5.0 ext{ m}$ and angular frequency $\omega ext{ [rad/s]}$:

$$x(t) = A \sin(\omega t), \quad y(t) = A \sin(2\omega t), \quad z(t) = -2.5 ext{ m}$$

Differentiating analytically with respect to time yields the peak kinematic demands:

- **Maximum Commanded Velocity:** $v_{max} = \sqrt{5} \cdot A \cdot \omega pprox 2.236 \cdot A \cdot \omega$
- **Maximum Commanded Acceleration:** $a_{max} = 4 \cdot A \cdot \omega^2$

Table 1 summarizes the commanded kinematics alongside the steady-state tracking error and payload swing metrics obtained across three randomized trials per rate.

Rate $ \omega$ (rad/s)	v_{max} (m/s)	a_{max} (m/s²)	Baseline 3D Error (m)	Geometric 3D Error (m)	Tracking Improvement	Mean Swing Angle
0.15	1.68	0.45	0.179	0.220	-23.1%	20.90°
0.20	2.24	0.80	0.283	0.313	-10.6%	26.17°
0.25	2.80	1.25	0.446	0.339	**+24.0%**	31.15°
0.30	3.35	1.80	0.676	0.384	**+43.2%**	36.07°

Key Analytical Findings:
1. Low-Speed Regime ($\omega \le 0.20 ext{ rad/s}$): At $\omega = 0.15$, required acceleration is very modest ($a_{max} = 0.45 ext{ m/s}^2$). The fact that the baseline PX4 controller outperforms the geometric controller here demonstrates a **controller tuning / bandwidth trade-off**. The geometric attitude gains were optimized for strong dynamic disturbance rejection, causing mild over-correction in slow, quasi-static conditions.

2. High-Speed Regime ($\omega \ge 0.25 ext{ rad/s}$): Because commanded acceleration scales quadratically ($\propto \omega^2$), a_{max} reaches $1.80 ext{ m/s}^2$ at $\omega = 0.30$. In this regime, tracking errors grow across both architectures due to **trajectory aggressiveness** pushing against physical rotor thrust and vehicle tilt bounds. However, the geometric controller maintains higher effective bandwidth under heavy dynamic coupling, delivering a **43.2% error reduction** and a **12.9% reduction in payload swing** over the baseline.

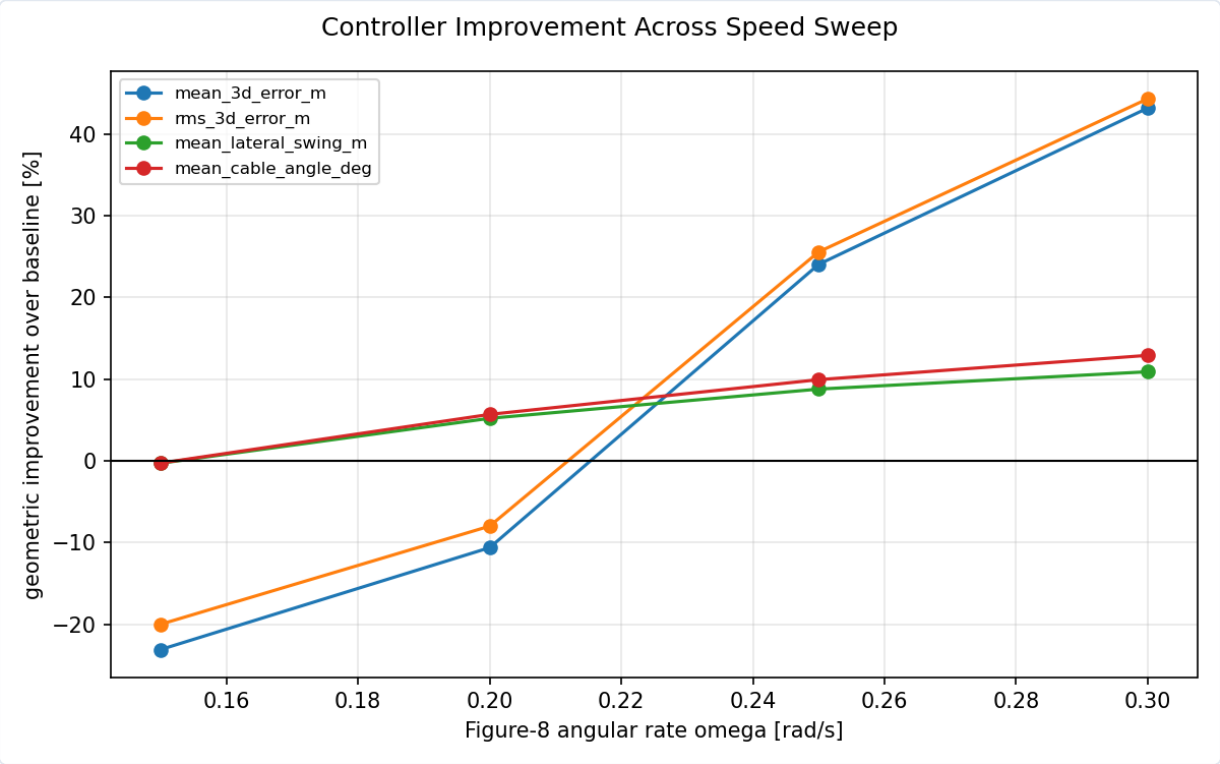


Figure 1: Percentage improvement of geometric controller over PX4 baseline across commanded trajectory rates. Note the crossover to positive advantage at $\omega \ge 0.23 ext{ rad/s}$.

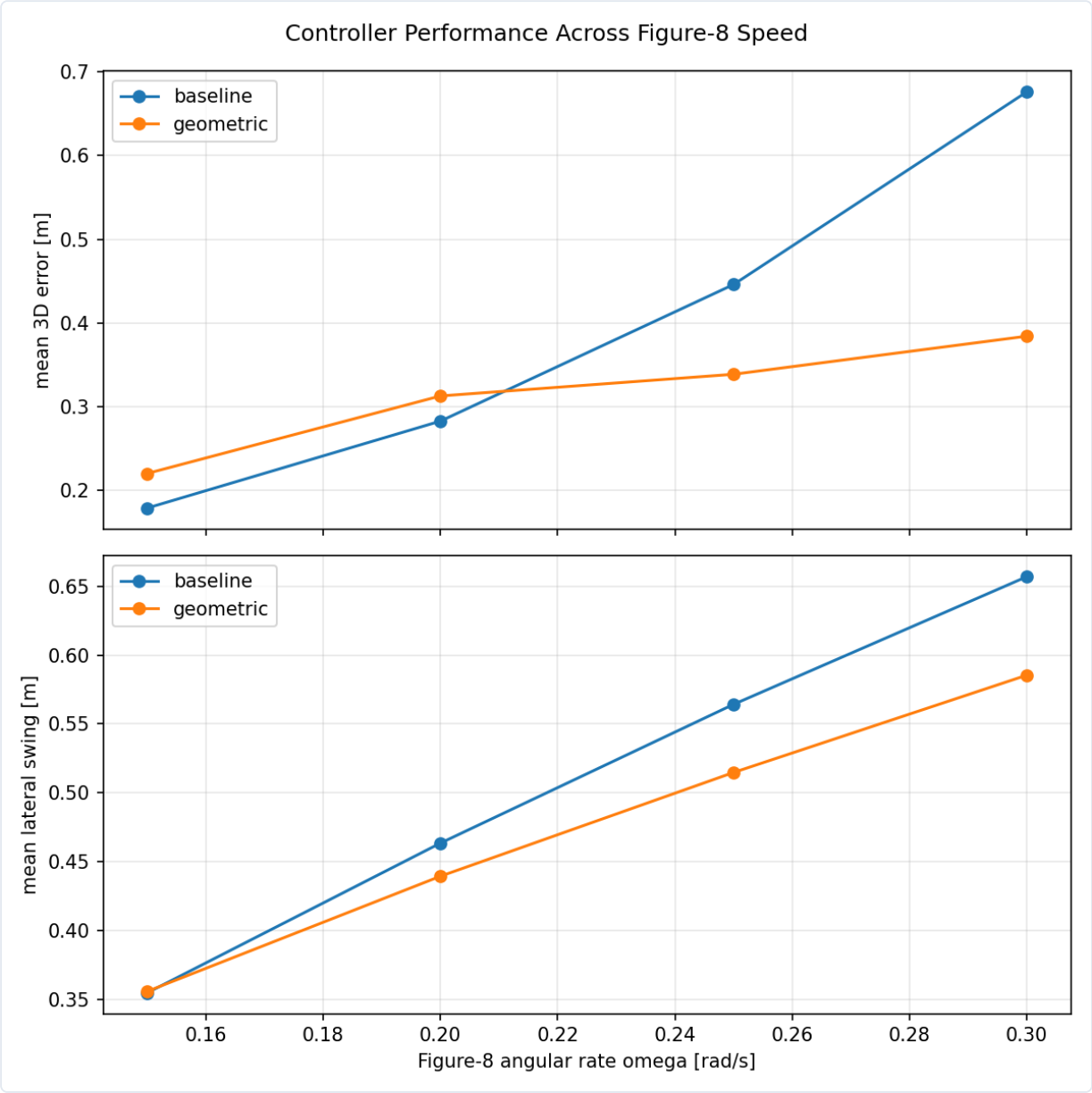


Figure 2: Absolute 3D position tracking error (left) and cable swing angle (right) as a function of trajectory rate ω.

3. Software Environment Reproducibility

To ensure seamless reproduction for student researchers without dependency conflicts, the repository provides fully pinned environments and automated testing.

3.1 Pinned Dependency Version Matrix

Component	Pinned Version / Branch	Role in System
ROS 2	Humble Hawksbill (Desktop)	Middleware, high-level geometric control & telemetry logging
Gazebo	Classic 11.10.2	ODE physics engine (250 Hz step), slung-payload dynamics simulation
PX4 Autopilot	v1.15.2 (git tag)	SITL autopilot firmware, EKF2 state estimation
Micro XRCE-DDS Agent	v2.4.2 (git tag)	High-throughput client-to-agent bridge between PX4 uORB and ROS 2
px4_msgs	release/1.15 (git branch)	Interface definitions for ROS 2 offboard control modes and setpoints

3.2 One-Command Reference Reproduction

Starting from a clean Linux machine with Docker installed, executing the following two commands automatically compiles and evaluates the reference Figure-8 trial:

```
# 1. Clone repository and build Docker image
git clone https://github.com/paneendrakumar0/uav-autonomous-telemetry.git
cd uav-autonomous-telemetry
docker build -t uav-payload-sitl .

# 2. Run the automated reference test (headless)
docker run --rm -v $(pwd):/workspaces/uav-autonomous-telemetry \
  -w /workspaces/uav-autonomous-telemetry uav-payload-sitl ./run_reference_test.sh
```

3.3 Pre-Defined Acceptance Tolerances & Empirical Results

The reference script (`run_reference_test.sh`) launches PX4 SITL and the ROS 2 controller at $\omega = 0.25 ext{ rad/s}$. Once steady-state flight is reached ($t \ge 25 ext{ s}$), the test evaluates position error against a strict acceptance gate:

```
========================================
UAV Slung-Payload Automated Reference Test
========================================
[1/4] Starting Micro XRCE-DDS Agent...
[2/4] Starting PX4 SITL (Headless)...
[3/4] Launching ROS 2 offboard control (omega=0.25)...
Waiting 85 seconds for experiment to collect steady-state data...
[4/4] Cleaning up processes...
========================================
Analyzing Metrics
========================================
Post-25s Mean 3D Tracking Error: 0.406 m
[PASS] Error 0.406 m is < 0.50 m threshold.
```

4. Current Research Progress: Payload-Aware Convex MPC

Following these reproducibility improvements, research has advanced to 3D obstacle avoidance using a **Payload-Aware Convex Model Predictive Controller (MPC)**:

- **Coupled Dynamics:** Linearized 10-state space (A_d, B_d) capturing quadrotor kinematics and pendulum swing ($\ddot{ heta} pprox -(g/L) heta - (1/L)a_{xy}$).

- **Safety Damping:** Strict constraint on payload swing angle ($ heta_{swing} \le 15^\circ$) during evasive maneuvers.
- **Benchmark World:** Integrated realistic asphalt runway, 5m cylindrical pillars, 4m wide gap gate, and terminal deceleration barrier.
- **Experimental Validation:** Obstacle avoidance achieved with **1.59m clearance** at Pillar 1 and peak payload swing dampened to **4.15°** (well below the 15° threshold).